Grade 8
Unit 14
Lesson 1 Practice

Name_Answer Key________
Date___________________
Period_________________

Brielle has a deck of a cards that she separates by suits. She picks up the Ace through the King of hearts. Suppose Brielle randomly chooses a card from the pile of hearts, replaces it, and then chooses a second card.

1. List the sample space of picking a card out of the pile of hearts, replacing the first card after it's been picked. $A = $ Ace, $J = $ Jack, $Q = $ Queen, $K = $ King

$$\{AA, A2, A3, A4, A5, A6, A7, A8, A9, A10, AJ, AQ, AK,\\ 2A, 2\,2, 2\,3, 2\,4, 2\,5, 2\,6, 2\,7, 2\,8, 2\,9, 2\,10, 2J, 2Q, 2K,\\ 3A, 3\,2, 3\,3, 3\,4, 3\,5, 3\,6, 3\,7, 3\,8, 3\,9, 3\,10, 3J, 3Q, 3K,\\ 4A, 4\,2, 4\,3, 4\,4, 4\,5, 4\,6, 4\,7, 4\,8, 4\,9, 4\,10, 4J, 4Q, 4K,\\ 5A, 5\,2, 5\,3, 5\,4, 5\,5, 5\,6, 5\,7, 5\,8, 5\,9, 5\,10, 5J, 5Q, 5K,\\ 6A, 6\,2, 6\,3, 6\,4, 6\,5, 6\,6, 6\,7, 6\,8, 6\,9, 6\,10, 6J, 6Q, 6K,\\ 7A, 7\,2, 7\,3, 7\,4, 7\,5, 7\,6, 7\,7, 7\,8, 7\,9, 7\,10, 7J, 7Q, 7K,\\ 8A, 8\,2, 8\,3, 8\,4, 8\,5, 8\,6, 8\,7, 8\,8, 8\,9, 8\,10, 8J, 8Q, 8K,\\ 9A, 9\,2, 9\,3, 9\,4, 9\,5, 9\,6, 9\,7, 9\,8, 9\,9, 9\,10, 9J, 9Q, 9K,\\ 10A, 10\,2, 10\,3, 10\,4, 10\,5, 10\,6, 10\,7, 10\,8, 10\,9, 10\,10, 10J, 10Q, 10K,\\ JA, J2, J3, J4, J5, J6, J7, J8, J9, J10, JJ, JQ, JK,\\ QA, Q2, Q3, Q4, Q5, Q6, Q7, Q8, Q9, Q10, QJ, QQ, QK,\\ KA, K2, K3, K4, K5, K6, K7, K8, K9, K10, KJ, KQ, KK\}$$

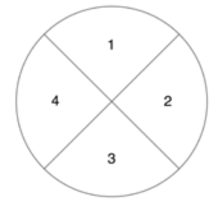
2. Use the sample space to determine how likely it is that Brielle will choose the Queen of hearts both times. Create a ratio to compare the number of outcomes with only the Queen of Hearts to the total number of outcomes in the sample space.

$$\frac{1}{169}$$

3. Predict how the sample space and the probability of choosing the Queen of hearts each time would change if Brielle uses the whole deck of cards and not just the hearts.

The probability would become even smaller because the sample space would get much larger.

Suppose a fair spinner numbered 1 through 4 is spun 2 times. Complete the tree diagram showing all possible outcomes of spinning the spinner twice. On the first spin, the options are $1, 2, 3$, and 4. On the second spin, the options are $1, 2, 3$, and 4.



4. Write the possible outcomes for the first spin.

   $\{1, 2, 3, 4\}$

5. Write the possible outcomes for the second spin.

   $\{1, 2, 3, 4\}$

6. Write the sample space.

First Spin: 1 2 3 4

Second Spin:

1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4

Sample Space: $\{11, 12, 13, 14, 21, 22, 23, 24, 31, 32, 33, 34, 41, 42, 43, 44\}$

7. Complete the table.

Experiment	Number of Outcomes in the First Step	Number of Outcomes in the Second Step	Total Number of Outcomes in the Sample Space
Spinning a fair spinner (numbered $1-4$) twice	4	4	16

8. Create a tree diagram to determine the sample space of flipping a fair coin four times.

H T — 1st flip

H T H T — 2nd flip

H T H T H T H T — 3rd flip

H T H T H T H T H T H T H T H T — 4th flip

9. How many events are in Event A = Flipping all tails?

   $\frac{1}{16}$

10. How many events are in Event B = Flipping exactly one tail?

   $\frac{4}{16}$ or $\frac{1}{4}$ $(\text{HHHT}, \text{HHTH}, \text{HTHH}, \text{THHH})$